Heterogeneous Privacy-Preserving Blockchain-Enabled Federated Learning for Social Fintech

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***Abstract*—Social fintech integrates financial technology with social networking to enhance financial services' accessibility and personalization by leveraging social interactions and user data. This approach raises privacy security concerns, particularly in application based on centralized artificial intelligence systems. To address these issues, blockchain-enabled federated learning (BEFL) offers a decentralized solution, improving robustness and privacy but facing challenges such as privacy attacks and heterogeneous crypto system. In response, a novel PKI and identity-based heterogeneous authenticated asymmetric group key agreement (PKI-IB-HAAGKA) protocol was proposed, which resolves crypto system heterogeneity issues. What's more, a PKI and identity-based heterogeneous batch multisignature (PKI-IB-HBMS) was proposed as a building block of PKI-IB-HAAGKA. This article presents the heterogeneous privacy-preserving blockchain-enabled federated learning (HPP-BEFL) system, designed to enhance privacy, security, and efficiency in social fintech applications. It effectively mitigates man-in-the-middle and inference attacks while improving overall system performance. Through security analysis and experiment results, it is demonstrated that the proposed PKI-IB-HAAGKA, PKI-IB-HBMS, and HPP-BEFL are provably secure and highly efficient, which can be applied to large-scale heterogeneous privacy-preserving model training scenarios.**

***Index Terms*—Blockchain-enabled federated learning (BEFL), heterogeneous asymmetric group key agreement (HBMS), inference attack, privacy preserving, social fintech.**

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I. Introduction

SOCIAL fintech is a transformative approach that integrates fintech [1] with social networking elements to enhance the accessibility and personalization of financial services. By leveraging social interactions and user behavior data from social platforms, social fintech provides customized financial products and services that foster trust and collaboration among users. This framework not only aims to optimize financial transactions but also to promote social good, aligning with the principles of social finance [2], which focuses on generating social and financial returns. Therefore, social fintech serves as a critical enabler of social finance, utilizing technology to improve engagement and create positive social impacts while addressing the financial needs of communities.

To provide customized financial products and services, social fintech heavily relies on artificial intelligence (AI) [3] technologies as well as vast amounts of user data. In conventional centralized learning frameworks, user data is kept on central servers, which can result in single point of failure (SPF) and risks of privacy leakage. The emergence of blockchain-enabled federated learning (BEFL) [4] has effectively addressed these issues, enhancing the robustness and privacy-preserving of training systems, and further accelerating the implementation of social fintech. In a blockchain-enabled federated learning paradigm, multiple users employ local devices, which store their data, to train a universal model in distributed manner. The parameters of these local models rather than the raw data are then forwarded to a central server for global model aggregation. A group of blockchain nodes manages the aggregation tasks. According to predefined policies, a set of blockchain nodes could be selected to perform the aggregation task in the federated learning based on the smart contract. Finally, the smart contract packages the newly trained global model into a block, which is verified and added to the blockchain through a consensus algorithm. This iterative process will be repeated multiple times until the machine learning model's accuracy reaches convergence.

As model attack techniques continue to evolve, malicious attackers may intercept unprocessed model parameters during the training process via network sniffing and utilize inference attacks [5] to obtain users' private data. To counter inference attacks, cryptographic technologies have been

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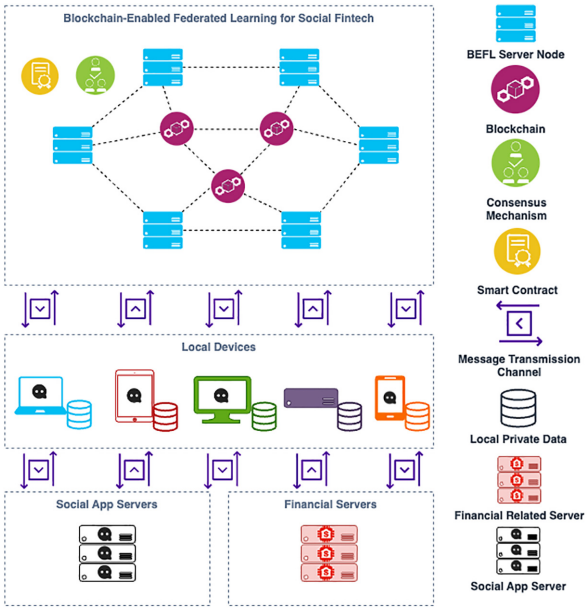


Fig. 1. Blockchain-enabled federated learning for social fintech.

introduced into the blockchain-enabled federated learning paradigm. Common cryptographic algorithms that protect privacy include homomorphic [6] and secure multiparty computation [7]. Homomorphic encryption allows local models to be encrypted before being uploaded, enabling the central server to aggregate the models without decrypting them. However, the complexity, high computational demands, and large storage requirements of homomorphic encryption can reduce training efficiency while increasing the storage and computational burden on the central aggregation server. Secure multiparty computation allows multiple computing devices to protect model information by adding masks to local models, with the masks being canceled out during aggregation by the central server. Multiple rounds of communication between local devices of secure multiparty computation may reduce training efficiency.

A promising approach is to introduce an efficient group key agreement [8] protocol into the blockchain-enabled federated learning paradigm, producing lightweight and efficient encryption and decryption keys. All nodes in BEFL participate in the group key agreement, resulting in a single public encryption key and an individual decryption key for each node. In this way, each local computing device could upload local training model encrypted with the public encryption key which only could be decrypted by the BEFL nodes. However, traditional group key agreement protocols generate only symmetric keys, leading to a linear increase in the number of keys as participants grow, which complicates key management. Moreover, traditional key agreement protocols do not account for the heterogeneous nature of blockchain nodes in the federated learning scenario, where this article mainly considers the heterogeneity of cryptographic systems.

Inspired by this, the article proposes the PKI and identity-based heterogeneous batch multisignature (PKI-IB-HBMS) protocol and PKI and identity-based heterogeneous authenticated asymmetric group key agreement (PKI-IB-HAAGKA) protocol, which effectively addresses the key management and participant heterogeneity issues unsolved by traditional group key agreement protocols. This solution enables participants from heterogeneous cryptographic systems, such as PKI and identity-based systems, to jointly negotiate a publicly shared encryption key while each participant retains a private decryption key. Anyone with access to the participants' public key set can encrypt messages using the shared encryption key, but only those with the decryption key can decrypt them. Based on PKI-IB-HAAGKA as a fundamental component, this article proposes the heterogeneous and privacy-preserving blockchain-enabled federated learning (HPP-BEFL) system, which effectively resolves both inference attacks and cryptographic heterogeneity in the current blockchain-enabled federated learning paradigm.

The primary contributions of this article are as follows.

1) This article introduces an innovative PKI-IB-HBMS solution and demonstrates its security. Building on this foundation, proposes the PKI-IB-HAAGKA solution. This new solution addresses key management and participant heterogeneity issues found in traditional key agreement protocols. The security of the proposed PKI-IB-HAAGKA could also be proved What's more, through theoretical analysis, it's proved that both of the proposed building blocks achieve linear communication complexity.
2) Utilizing PKI-IB-HAAGKA as a fundamental component, this article proposes the HPP-BEFL system, which effectively resolves inference attacks and cryptographic heterogeneity in traditional blockchain-enabled federated learning systems. This article also demonstrates the security of this system.
3) This article presents a series of efficiency experiments demonstrating that the proposed PKI-IB-HAAGKA outperforms traditional cryptographic methods, such as heterogeneous encryption (HE) and secure multiparty computation (SMPC). Additionally, extensive experiments with real-world datasets such as MNIST [9] and CIFAR-10 [10] demonstrate that the HPP-BEFL system is both highly efficient and secure against inference and man-in-the-middle attacks.

The remaining sections of this article are structured as follows: Section II introduces the necessary background knowledge, including federated learning, blockchain, asymmetric group key agreement protocols, and bilinear mapping. Section III discusses the related works of privacy-preserving federated learning, blockchain-enabled federated learning, asymmetric group key agreement protocols. Section IV details the basic components of the HPP-BEFL solution, including the concrete constructions and security analysis of PKI-IB-HAAGKA and PKI-IB-HBMS. Section V describes the system model, threat model, and concrete constructions of the proposed HPP-BEFL system. Section VI presents the functional analysis, security analysis, and performance evaluation of HPP-BEFL. Section VII summarizes the proposed article and discusses the future works.

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II. Preliminary

A. Blockchain

Blockchain technology is a decentralized digital ledger that provides secure, transparent, and tamper-proof record-keeping, originally developed as the foundation for cryptocurrencies such as Bitcoin [11]. Essentially, a blockchain is composed of a series of blocks, each containing a list of transactions. Each block is connected to its predecessor via cryptographic hashes, forming a chronological and unchangeable record. This structure guarantees that once data is recorded, it cannot be modified without the consensus of network participants, thereby strengthening data integrity. The underlining mechanisms such as consensus mechanisms [12], smart contracts [13], cryptography [14] safeguards the blockchain system's overall security and privacy.

B. Federated Learning

Federated learning (FL) [15] is an innovative machine learning approach that facilitates decentralized model training across multiple local devices, allowing them to use local data without sharing private information. This paradigm addresses key challenges in traditional machine learning, particularly concerning data privacy, security, and regulatory compliance. In federated learning, each local device trains a model using its own private dataset and shares only model updates, such as parameters or weights, with a central aggregation server. The server aggregates these updates to enhance a global model, which is then redistributed to the devices for further training. This iterative process repeats until the model achieves satisfactory performance.

C. Asymmetric Group Key Agreement

The AGKA protocol [16] establishes a secure and credentialed channel among a group of participants using a commonly accessible encryption key ek and a decryption key set $\{dk_i\}$ for each participant, with each decryption key generated solely by the respective protocol participant. An AGKA protocol may include the following algorithms.

1) *Setup*: When provided ℓ as a secure parameter, generate the public parameter list pp.
2) *Extract*: Generate public and private key (pk_i, sk_i) pair for each participant.
3) *Agreement*: All participants work together to generate a set of encryption key slices and decryption key slices for future key generation.
4) *Enc.Gen*: Generate the commonly accessible encryption key ek with the encryption key slices produced by agreement step.
5) *Dec.Gen*: Each participant generates his/her own private decryption key dk_i with the decryption key slices produced by agreement step.
6) *Enc*: Given a plaintext m, anyone could encrypt m with commonly accessible encryption key ek, which results into a ciphertext c.
7) *Dec*: Given a ciphertext c which has been encrypted with the commonly accessible encryption key ek, protocol participants could decrypt c with self-produced decryption key dk_i into plaintext m.

D. Symmetric Bilinear Mapping and Computational Assumption

Let $\mathbb{G}_1$, $\mathbb{G}_2$ and $\mathbb{G}_T$ be three multiplicative groups of prime order q, and u be a generator of $\mathbb{G}_1$, v be a generator of $\mathbb{G}_2$. A map $\hat{e}: \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ is called a bilinear mapping [17] if it satisfies the following properties.

1) *Bilinearity*: $\hat{e}(u^a, v^b) = \hat{e}(u, v)^{ab}$ for all $a, b \in \mathbb{Z}_q^\star$
2) *Nondegeneracy*: there exists $u \in \mathbb{G}_1, v \in \mathbb{G}_2$ such that $\hat{e}(u, v) \neq 1$.
3) *Computability*: there exists an efficient algorithm to compute $\hat{e}(u, v)$ for any $u \in \mathbb{G}_1, v \in \mathbb{G}_1$,

When $\mathbb{G}_1 = \mathbb{G}_2$, the special bilinear mapping is called as symmetric bilinear mapping, which will be an fundamental component of the proposed building block.

The security of the proposed PKI-IB-HBMS and PKI-IB-HAAGKA are based on the CDH problem and k-BDHE problem which are assumed to be hard.

Definition 1: CDH Problem. given g, g^a, g^b for unknown $a, b \in \mathbb{Z}_q^\star$, compute g^{ab}.

Definition 2: CDH Assumption. let $\mathcal{A}$ be an algorithm which has advantage

$$\mathrm{Adv}(\mathcal{A}) = Pr[\mathcal{A}(g, g^a, g^b) = g^{ab}]$$

in solving the *CDH problem*. The CDH assumption states that $\mathrm{Adv}(\mathcal{A})$ is negligible for any polynomial-time algorithm $\mathcal{A}$.

Definition 3: k-BDHE Problem. given g, h, and $y_i = g^{\alpha^i}$ in $\mathbb{G}_1$ for $i = 1, 2, \ldots, k, k+2, \ldots, 2k$ as input, compute $\hat{e}(g, h)^{\alpha^{k+1}}$.

Definition 4: k-BDHE Assumption. let $\mathcal{A}$ be an algorithm which has advantage $\mathrm{Adv}(\mathcal{A})$

$$Pr[\mathcal{A}(g, h, y_1, \ldots, y_k, y_{k+2}, \ldots, y_{2k}) = \hat{e}(g, h)^{\alpha^{k+1}}]$$

in solving k-BDHE problem. The k-BDHE assumption states that $\mathrm{Adv}(\mathcal{A})$ is negligible for any polynomial-time algorithm $\mathcal{A}$.

III. Related Works

A. Privacy-Preserving Federated Learning

While federated learning provides a framework for decentralized machine learning, enabling model training without the necessity of uploading local datasets, it does not fully guarantee privacy in real-world applications. In federated learning, numerous privacy-preserving strategies predominantly utilize the differential privacy (DP) [18] or the cryptographic methods [19] to safeguard user data. Various approaches to DP have been proposed by researchers. For example, Wei et al. [20] developed a privacy budget distribution strategy for both the internal and external update phases, utilizing Rényi differential privacy composition principles. Wang et al. [21] proposed a

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novel framework for privacy-preserving edge federated learning, utilizing local differential privacy and presenting three distinct mechanisms to address privacy challenges during the federated learning process. More recently, the FedFIM framework was introduced by [22], integrating a differential privacy mechanism into its architecture.

In the realm of homomorphic encryption (HE), the E-PEFL scheme is novelly proposed by [23], designed to shield from inference and poisoning attacks using HE techniques. Building on this, BatchCrypt scheme was introduced by [24], which facilitates model training while keeping local parameter information confidential. Within the framework of secure multiparty computation (SMC), a centralized federated learning approach was developed by [25], employing SMC to protect the privacy of local parameters among various user groups. Additionally, PrivacyFL was introduced by [26], a scheme that integrates SMC and DP to strengthen security by combining both technologies. Ma et al. [27] proposed a federated learning framework enhanced by blockchain, designed to mitigate potential Byzantine attacks.

B. Blockchain-Enabled Federated Learning

FL has appeared as a viable approach for facilitating distributed machine learning coupled with privacy-preserving, as it allows local devices to train models with local data while avoiding the necessity of sharing it with a central server. Nevertheless, traditional FL encounters several challenges, including a single point of failure, insufficient incentive mechanisms, and the risk of data falsification. To overcome these challenges, blockchain-enabled federated learning (BEFL) has been proposed, attracting considerable interest from both academia and industry."

Blockchain technology, known for its decentralized nature, has proven effective in providing privacy and security in various Internet of Things (IoT) scenarios [28]. By integrating blockchain into FL, a decentralized machine learning paradigm can be achieved, removing the requirement for a centralized server and mitigating risks such a man-in-the-middle attacks (MMA) and single-point failures [29]. The decentralized ledger and consensus mechanisms in blockchain ensure the integrity and transparency of model updates, making BEFL more robust against malicious actors.

Qu et al. [30] issued the privacy in BEFL and proposed a game-centric approach based on the Markov decision process to enhance privacy in industrial settings [31]. Furthermore, Li et al. [32] puts forward a unique consensus algorithm designed specifically for BEFL to boost efficiency. Researchers have also explored the application of BEFL in various domains, such as vehicular networks [33], [34], where the decentralized and secure nature of blockchain can offer significant benefits.

However, despite these advancements, BEFL still faces challenges. For example, Kim and Hong [35] were the first to utilize a proof-of-work (PoW)-based blockchain for FL, replacing the central server with a decentralized architecture. While this approach enhances security, it introduces communication overhead due to the end-to-end switch mechanism. To tackle this, Kim et al. [36] proposed sample reduction techniques on the local side to improve training efficiency.

C. AGKA

AGKA combines the features from GKA [8] and broadcast encryption (BE) [37], which allows a group of entities negotiated one public accessible encryption key and each entity's unique and private decryption key. The notion of AGKA was first proposed by [16]. This protocol only addresses passive attacker scenarios. Authenticated AGKA (AAGKA) protocols are intended to mitigate threats from active attackers, such as those found in Public Key Infrastructure (PKI) crypto system [38], Identity-Based crypto system [39], [40] and certificateless public key crypto system [41], [42]. Despite this, none of the above mentioned protocols are extensively used in practice, for all these systems remain static. Thus, dynamic AAGKA protocols for Identity-Based crypto system have been proposed [43], [44], but they limit the group size to the initial number of members. An AAGKA protocol incorporating sender nonrepudiation and privacy (AAGKAwSNP) [45] was developed for identity-based crypto system, yet it only supports static group communications. Zhang et al. [46] presented a dynamic authenticated asymmetric group key agreement protocol DAAGKAwSNP that ensures sender nonrepudiation and privacy for secure group communication in real-time applications.

IV. Building Block

In this section, this article introduces the PKI and identity-based heterogeneous batch multisignature (PKI-IB-HBMS) protocol as well as the PKI and identity-based heterogeneous authenticated asymmetric group key agreement (PKI-IB-HAAGKA) protocol, both of which serve as foundational components for the proposed heterogeneous privacy-preserving blockchain-enabled federated learning (HPP-BEFL) system. For simplicity and readability, the notations and definitions of the building block are given in Appendix A.

A. PKI-IB-HBMS

The PKI-IB-HBMS is the building block of the proposed PKI-IB-HAAGKA. The concrete construction and security analysis of the proposed PKI-IB-HBMS will be discussed in this part.

1) Concrete Construction: In the proposed PKI-IB-HBMS, each participants with unique index from PKI or identity-based crypto system could produce t signatures for t distinct messages along with a random value, resulting in a $t+1$ element batch signature. What's more, all generated signatures on the same message can be assembled into one PKI-IB-HBMS composed of $t+1$ group items. Any one could verify the validity of the PKI-IB-HBMS with the set of all the protocol participants' public key.

The concrete construction of PKI-IB-HBMS comes as follows.

a) Setup(ℓ): Given ℓ as the security parameter, the key generation center (KGC) and the certificate authority (CA)

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work together to choose $\mathbb{G}_1$ and $\mathbb{G}_2$ as two cyclic multiplicative groups with prime order q, where g is the generator of $\mathbb{G}_1$ and there is a bilinear map $\hat{e}:\mathbb{G}_1\times\mathbb{G}_1\to\mathbb{G}_2$. Then select two cryptography hash functions $H_1,H_2:\{0,1\}^\star\to\mathbb{G}_1$. And KGC from Identity-Based crypto system selects $\kappa\in\mathbb{Z}_q^\star$ haphazardly as the *master-secret* and makes $g_1=g^\kappa$. The system outputs the public parameter list $pp=(\mathbb{G}_1,\mathbb{G}_2,\hat{e},H_1,H_2,g,g_1)$

b) HBMS.PKI.Extract(): At this phase, each participant from PKI crypto system $\mathcal{U}_{\mathrm{PKI}i(1\le i\le n_{\mathrm{PKI}})}$ generates his/her public and private key pair and applies for a certificate from CA for his/her public key. The following steps outline the process.
   i) $\mathcal{U}_{\mathrm{PKI}_i}$ computes his/her own public and private key pair as $(pk_{\mathrm{PKI}_i}=\hat{e}(sk_{\mathrm{PKI}i},g),sk_{\mathrm{PKI}_i}=h_i)$ where $h_i\in\mathbb{G}_1/\{1\}$ is randomly selected.
   ii) Then $\mathcal{U}_{\mathrm{PKI}i}$ sends his/her own pk_{PKI_i} with extra identifying information to CA to verify the ownership of $pk_{\mathrm{PKI}i}$.
   iii) CA checks $pk_{\mathrm{PKI}i}$ with $\mathcal{U}_{\mathrm{PKI}i}$'s uploaded information. If the check passes, CA creates and returns a certificate cert_i back.

c) HBMS.ID.Extract(ID_i): At this phase, each participant from identity-based crypto system $\mathcal{U}_{ID_i(1\le i\le n_{ID})}$ sends his/her identity ID_i to KGC to request his/her private and public key pair. The following steps outline the process.
   i) $\mathcal{U}_{ID_i}$ sends request to KGC with his/her own identity ID_i.
   ii) Once KGC receives request with an identity ID_i from $\mathcal{U}_{ID_i}$. It computes $\mathcal{U}_{ID_i}$'s public and private key pair as $(pk_{ID_i}=\hat{e}(id_i,g_1),sk_{ID_i}=s_i)$ where $s_i=id_i^\kappa,id_i=H_1(ID_i)$.
   iii) KGC sends $\mathcal{U}_{ID_i}$'s public and private key pair (pk_{ID_i},sk_{ID_i}) back.

d) PKI-ID-Sign($\{m_i\},\{sk_{\mathrm{PKI}/ID_i}\}$): At this phase, t messages $m_1,\ldots,m_t$ will be signed with all participants private keys. The following steps for each participant are outlined as follows.
   i) Computes $r_{\mathrm{PKI}/ID_i}=g^{\eta_{\mathrm{PKI}/ID_i}},d_{\mathrm{PKI}/ID_{i,j}}=sk_{\mathrm{PKI}/ID_i}f_j^{\eta_{\mathrm{PKI}/ID_i}}$ for all messages $m_1,\ldots,m_t$, where $\eta_{\mathrm{PKI}/ID_i}\in\mathbb{Z}_q^\star$ was randomly selected and $f_j=H_2(m_j)$.
   ii) Produces $\mathcal{U}_{\mathrm{PKI}i}$'s batch signature with certification cert_i as $\sigma_{\mathrm{PKI}i}=(r_{\mathrm{PKI}i},\{d_{\mathrm{PKI}i,j}\}_{(1\le j\le t)},\mathrm{cert}_i)$
   iii) Produces U_{ID_i}'s batch signature as $\sigma_{ID_i}=(r_{ID_i},\{d_{ID_i,j}\}_{(1\le j\le t)})$
   iv) Publishes $\sigma_{\mathrm{PKI}/ID_i}$.

e) Aggregate($\{\sigma_{\mathrm{PKI}/ID_i}\}$): Once all certification $\{\mathrm{cert}_i\}$ of $\mathcal{U}_{\mathrm{PKI}i}$ are valid, all signatures $\{\sigma_{\mathrm{PKI}/ID_i}\}$ on t messages from heterogeneous signers can be aggregated into one PKI-IB-HBMS $(\omega,d_1,\ldots,d_t)$, where $\omega=\prod_{i=1}^{n_{\mathrm{PKI}}}r_{\mathrm{PKI}i}\prod_{i=1}^{n_{ID}}r_{ID_i},d_j=\prod_{i=1}^{n_{\mathrm{PKI}}}d_{\mathrm{PKI}i,j}\prod_{i=1}^{n_{ID}}d_{ID_i,j}$

f) Verify($\{d_i\},\omega$): The validity of the aggregated PKI-IB-HBMS $(\omega,d_1,\ldots,d_t)$ could be checked by computing $Q=\prod_{i=1}^{n_{\mathrm{PKI}}}pk_{\mathrm{PKI}i}\prod_{i=1}^{n_{ID}}pk_{ID_i}$ with all participants' public key $pk_{\mathrm{PKI}/ID_i1\le i\le n}$ and checking all the equations $\hat{e}(f_j,\omega)\cdot Q\stackrel{?}{=}\hat{e}(d_j,g)_{\{1\le j\le t\}}$ If all the equations are equal, the PKI-ID-HBMS is valid; otherwise, an invalid index set $\mathbb{I}$ could be produced.

2) Security Analysis: The security model of the proposed PKI-IB-HBMS is given in Appendix C.

Theorem 1: In the setting of the random oracle paradigm and the security framework described by adaptive chosen-message attacks [47], the proposed PKI-IB-HBMS's existential unforgeability can be deduced from the CDH assumption.

Proof: The proof of Theorem 1 is given in Appendix D.

B. PKI-IB-HAAGKA

In the proposed PKI-IB-HAAGKA, there are two types of polynomially bounded number of potential participants from two different crypto system, the public key infrastructure (PKI) crypto system and the identity-based crypto system.

Let the set of participants from PKI crypto system be $\mathbb{U}_{\mathrm{PKI}}$ and the set from ID-based crypto system be $\mathbb{U}_{ID}$. Anyone from PKI crypto system holds his/her own unique index i and a pair of public key and private key $(pk_{\mathrm{PKI}_i},sk_{\mathrm{PKI}_i})$ authorized by CA (certificate authority). And anyone from $\mathbb{U}_{ID}$ holds his/her own Identity ID_i, a unique index i and a secret key s_i.

Any subset $\mathbb{U}\subseteq\mathbb{U}_{PKI}\cup\mathbb{U}_{ID}$ could determine to construct a safe communication channel at any time. In this article, $\Omega_{\mathcal{U}_i}^{\chi}$ represents instance χ of participant $\mathcal{U}_i$.

An instance stops forwarding and collecting messages as long as it's *successfully terminated*. $\Omega_{\mathcal{U}_i}^{\chi}$ terminates or accepts the PKI-IB-AAGKA protocol when possesses $(ek_{\mathcal{U}_i}^{\chi},dk_{\mathcal{U}_i}^{\chi},\mathrm{pid}_{\mathcal{U}_i}^{\chi},\mathrm{sid}_{\mathcal{U}_i}^{\chi})$ where $ek_{\mathcal{U}_i}^{\chi}\neq\mathrm{null}$ and $dk_{\mathcal{U}_i}^{\chi}\neq\mathrm{null}$.

1) Concrete Construction: The concrete construction of PKI-IB-HAAGKA comes as follows.

a) Setup(ℓ): Based on HBMS.Setup, this stage selects extra identity-based signature scheme, PKI signature scheme, and another hash function $H_3:\mathbb{G}_2\to\{0,1\}^\zeta$, where ζ stands for the bit-size of the plaintext.

b) PKI.Extract(): The same as HBMS.PKI.Extract.

c) ID.Extract(ID_i): The same as HBMS.ID.Extract.

d) PKI-ID-Agreement(sk_{PKI/ID_i}): Each participant $\mathcal{U}_{\mathrm{PKI}/ID_i}$ performs the following steps.
   i) Randomly chooses $\eta_{\mathrm{PKI}/ID_i}\in Z_q^\star$ and computes $r_{\mathrm{PKI}/ID_i}=g^{\eta_{\mathrm{PKI}/ID_i}}$
   ii) Computes $f_j=H_2(j),d_{\mathrm{PKI}/ID_{i,j}}=sk_{i_{\mathrm{PKI}/ID}}f_j^{\eta_{\mathrm{PKI}/ID_i}}$ where $1\le j\le n$.
   iii) Makes participant $\mathcal{U}_{\mathrm{PKI}i}$'s message with certification cert_i generated by CA as $\mathbb{M}_{\mathrm{PKI}i}=(r_{\mathrm{PKI}i},\rho_{\mathrm{PKI}i},d_{\mathrm{PKI}i,j},\mathrm{cert}_i)$ where $1\le j\le n_{\mathrm{PKI}},j\neq i$ and ρ_{PKI_i} is the PKI signature on $r_{\mathrm{PKI}i}$ with private key $sk_{\mathrm{PKI}i}$.
   iv) Makes participant $\mathcal{U}_{ID_i}$'s message as $\mathbb{M}_{ID_i}=(r_{ID_i},\rho_{ID_i},d_{IDi,j})$ where $1\le j\le n_{ID},j\neq i$ and ρ_{ID_i} is the identity-based signature on r_{ID_i} with private key sk_{ID_i}.
   v) Publishes $\mathbb{M}_{PKI_i}$ and $\mathbb{M}_{ID_i}$.

e) Enc.Key.Gen($\{\mathbb{M}_{\mathrm{PKI}/ID_i}\}$): To generate the public accessible group encryption key, each participant checks the

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validity of all the signature-message pairs $\{(\rho_{\mathrm{PKI}/ID_i}, r_{\mathrm{PKI}/ID_i})\}$ as well as all the certification $\{\mathrm{cert}_i\}$. If the check passes, then computes $\omega = \prod_{i=1}^{n_{\mathrm{PKI}}} r_{\mathrm{PKI}i} \prod_{i=1}^{n_{ID}} r_{IDi}, Q = \prod_{i=1}^{n_{\mathrm{PKI}}} pk_{\mathrm{PKI}i} \prod_{i=1}^{n_{ID}} pk_{IDi}$ and sets the public accessible encryption key as $ek = (\omega, Q)$.

f) Dec.Key.Gen$(\{\mathbb{M}_{\mathrm{PKI}/ID_i}\})$: Any $\mathcal{U}_{\mathrm{PKI}/ID_i}$ could check the validity of all signature-message pairs $\{(\rho_{\mathrm{PKI}/ID_i}, r_{\mathrm{PKI}/ID_i})\}$ and all the certification $\{\mathrm{cert}_i\}$. If all of these checks pass, then the participant could compute his/her own decryption key as: $dk_i = \prod_{j=1}^{n_{\mathrm{PKI}}} d_{\mathrm{PKI}j,i} \prod_{j=1}^{n_{ID}} d_{IDj,i}$. If the equation $Q \cdot \hat{e}(f_i, w) \stackrel{?}{=} \hat{e}(dk_i, g)_{1 \le i \le n}$ holds, $\mathcal{U}_{\mathrm{PKI}/ID_i}$ accepts dk_i as his/her own decryption key; otherwise, abort.

g) Enc(m, ek): Given a plaintext m, any one could produce the ciphertext with ek in the following steps:

i) Randomly picks $e \in Z_q^\star$, computes $c_1 = g^e, c_2 = \omega^e, c_3 = H_3(Q^e) \oplus m$

ii) Produces the ciphertext as $c = (c_1, c_2, c_3)$

h) Dec(c, dk_i): To decrypt the ciphertext $c = (c_1, c_2, c_3)$ with any $\mathcal{U}_{\mathrm{PKI}/ID_i}$'s decryption key dk_i, computes $m = c_3 \oplus H_3(\hat{e}(dk_i, c_1)\hat{e}(f_i^{-1}, c_2)))$.

The matrix below figuratively describes how the commonly accessible encryption key $ek = (\omega, Q)$ and the private decryption key set $\{dk_i\}$ are produced. For simplicity, $\mathcal{U}_i, dk_i, r_i, pk_i$ are assumed to either form the PKI or from the identity-based crypto system.

$$\begin{bmatrix} & \mathcal{U}_1 & \mathcal{U}_2 & \dots & \mathcal{U}_n & \mathrm{All} \\ \mathcal{U}_1 \Rightarrow & d_{1,1} & d_{1,2} & \dots & d_{1,n} & (r_1, pk_1) \\ \mathcal{U}_2 \Rightarrow & d_{2,1} & d_{2,2} & \dots & d_{2,n} & (r_2, pk_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathcal{U}_n \Rightarrow & d_{n,1} & d_{n,2} & \dots & d_{n,n} & (r_n, pk_n) \\ \mathrm{dk} & dk_1 & dk_2 & \dots & dk_n & (\omega, Q). \end{bmatrix}$$

2) Security Analysis: The security model of the proposed PKI-IB-HAAGKA is given in Appendix E.

Theorem 2: In the setting of the random oracle paradigm and the security framework described by Indistinguishable against Chosen Plaintext Attack (IND-CPA) [48], the proposed PKI-IB-HAAGKA's confidentiality can be deduced from the k-BDHE assumption.

Proof: The proof of Theorem 2 is given in Appendix F.

V. Heterogeneous and Privacy-Preserving Blockchain-Enabled Federated Learning

The system model, threat model, and concrete construction of the proposed HPP-BEFL system are given in this section.

A. System Model

This article proposed a heterogeneous and privacy-preserving model training framework Fig. 2. In this framework, BEFL assists in aggregating the uploaded local model to train the global model.

There are two kinds of entities, i.e., the local computing devices and BEFL system that build up the proposed BEFL training system. In the proposed system, BEFL system could execute FL algorithms at any time with a set of local computing devices to work together to train a general deep learning model in a confidential channel. Specifically, below is a detailed description of these two types of entities.

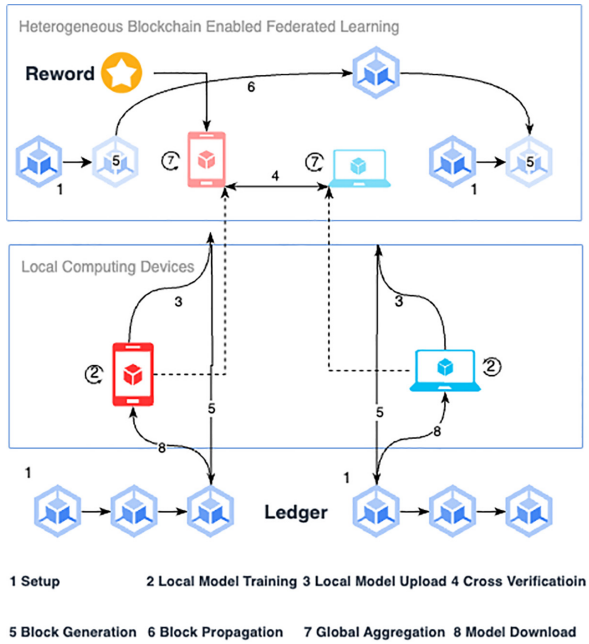


Fig. 2. System model.

1) *BEFL System*: BEFL system consists of a blockchain and a set of nodes defined as $\{\mathcal{N}_i\}$ from PKI or identity-based heterogeneous crypto system with an unique index i. At the system setup stage, all nodes will generate an initial global model W_g^0 and work together executing the proposed PKI-IB-HAAGKA to generate a global accessible encryption key ek and a set of decryption key $\{dk_i\}$ unique for each node. Then, $\mathcal{N}_i$ receives and decrypts the uploaded encrypted local models as $\{W_j^t\}$ to aggregate the global models W_g^t. What's more, all the nodes will store and record uploaded local models $\{W_j^t\}$ and global model W_g^t as a block into the blockchain.

2) *Local Devices* $\mathcal{U}_i$: Local training devices are defined as $\mathcal{U}_i$ with an unique index i. The local devices and the BEFL nodes $\{\mathcal{N}_j\}$ are bonded during setup stage. $\mathcal{U}_i$ trains local models with his/her own private datasets and uploads the trained local model (encrypted with ek) to the corresponding $\mathcal{N}_j$ to train general global models.

B. Threat Model

Definition 5: Man-in-the-Middle Attack.

An MMA refers to an attacker intercepting and tampering with the transmission of model updates and gradient information between participants and the central server. This allows the attackers to obtain sensitive data or influence the training results of the global model. Through such actions, the attackers can degrade model performance or leak participant privacy, compromising the system's security.

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TABLE I
SYMBOLS IN HPP-BEFL

Symbols	Meaning
$\mathcal{N}_i$	Heterogeneous BEFL node with unique index i
$\mathcal{U}_i$	Local device with unique index i
W_g^0	Initial global model
ek	Commonly accessible encryption key
dk_i	Unique decryption key of $\mathcal{N}_i$
n	Number of D_i elements
D_i	$\mathcal{U}_i$'s local private dataset
$\|D_i\|$	Size of D_i
x_i	Training sample
y_i	Label of x_i
$l(W; D_i)$	Loss function of $\mathcal{U}_i$
W_i^t	$\mathcal{U}_i$'s updated model in iteration t
W_g^{t-1}	$\mathcal{U}_i$'s last iteration's global model fetched from BEFL
W_g	Global model
ϵ	Learning rate
$\nabla l(W_g^{t-1}; D_i)$	Derivatives of $l(W_g^{t-1}; D_i)$
c_i	$\mathcal{U}_i$'s encrypted local model
$\mathcal{B}^t$	Generated block in interation t
W_g^t	General global model

Definition 6: Inference Attack.
The inference attack (IA) is a kind of privacy attack in which an adversary try best to dig out the sensitive privacy data from the model parameters by investigating the updates or outputs shared during the learning process. This can involve extracting details about individual data points or identifying patterns that compromise data privacy, exploiting the iterative updates exchanged between local devices and the server.

In the proposed system, all aggregation server nodes in BEFL system are considered to be honest. They assist in receiving and aggregating encrypted uploaded local models for training a general global model. Nevertheless, malicious attackers outside the BEFL environment are assumed to be curious about the local computing devices' privacy information and aim to decrease the global model performance. The malicious attackers may try to get uploaded local model during training process through the MMA and utilize inference attacks to dig out the privacy data.

C. Concrete Construction

In this section, the concrete construction of the proposed HPP-BEFL is as follows. For readability and simplicity the symbols in concrete construction of the proposed HPP-BEFL are listed in Table I.

The aim of the proposed HPP-BEFL is to train a global model W_g with uploaded local model parameters in a privacy-preserving way. In the procedure of training the general global model, imperfect model's parameters are continuously revised to mitigate the gap between the true label y_j and the prediction label $\hat{y}_j$.

The overall workflow can be detailed as follows.

$$L(W) = \sum_{i=1}^{N} \frac{|D_i|}{|D|} l(W; D_i), D = \sum_{i=1}^{N} D_i.$$

It is crucial to highlight that attackers nowadays could exploit the transmitted model parameters information in the federated learning process to infer or reconstruct the original training data by inference attack which may have huge impact on participants' privacy. A encryption scheme mitigates this problem by sending the encrypted local parameters. In the proposed BEFL system, all aggregation server nodes will execute the proposed PKI-IB-HAAGKA first to generate a commonly accessible encryption key ek and decryption key set $\{dk_i\}$ unique for each node and select a aggregation server by a consensus algorithm at the setup stage. After BEFL setup finished, all local computing devices could encrypted their local training model by the BEFL public encryption key ek and upload local models to aggregation server in a privacy-preserving way. The BEFL node could decrypted the uploaded encrypted models and train a global model. What's more, the generated public encryption key ek and the final global model W_g^t will be recorded in the block. By this means, choice-making won't be regulated by a sole authority, assisting in preventing the misuse of authority and unethical manipulation.

The detailed construction comes as follows.

1) *Setup*: A preestablished function generates system parameters and initial global model W_g^0 which is distributed uniformly to all BEFL nodes. Then all the heterogeneous BEFL nodes $\{\mathcal{N}_i\}$ take part in the proposed PKI-IB-HAAGKA protocol which results in one commonly accessible encryption key ek which will be recorded on blockchain and decryption key dk_i unique for each node.
2) *Local Model Training*: Each local device $\mathcal{U}_i$ holds private local dataset $D_i = \{\langle x_j, y_j \rangle, j = 1, 2, \ldots, n\}$. To realize the previously mentioned aim, $\mathcal{U}_i$ utilizes the stochastic descent method [49] to train local model as $W_i^t = W_g^{t-1} + \epsilon \nabla l(W_g^{t-1}; D_i)$ for several predefined iterations.
3) *Local Model Upload*: Each $\mathcal{U}_i$ encrypts W_i^t with ek by the algorithm provided by the proposed PKI-IB-AAGKA.Enc: $c_i = \text{PKI-IB-AAGKA.Enc}(W_i^t, ek)$ and upload c_i to $\mathcal{N}_j$. Additionally, the associated local training time is packaged with c_i for validation.
4) *Cross Verification*: Once $\mathcal{N}_j$ receives $\{c_i\}$ from local devices, $\mathcal{N}_j$ decrypts all encrypted local models with its own decryption dk_j: $W_i^t = \text{PKI-IB-AAGKA.Dec}(c_i, dk_\star)$ $\mathcal{N}_j$ check the model parameters and processing time of the related devices in order. Determining if the computing time correlates with the data size influences the reliability of the model parameters. The validated data is kept in the prospective block of $\mathcal{N}_\star$ until checks pass and parameters are kept in a block.
5) *Block Generation*: $\mathcal{N}_j$ generates a candidate block $\mathcal{B}^t$.
6) *Block Propagation*: $\mathcal{N}_j$ broadcasts the $\mathcal{B}^t$ to all nodes $\{\mathcal{N}_i\}$. Every node in the system incorporates the new block into their local ledgers upon verification.
7) *Global Aggregation*: $\mathcal{N}_j$ executes the smart contract managed in BEFL system trains the global model with local models as $W_g^t = \sum_{i=1}^{N} (|D_i|/|D|) W_i^t$ which will be recorded into $\mathcal{B}^t$.
8) *Model Download*: All local devices could fetch the latest global model W_g^t and check the performance to decide whether continue the process.

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TABLE II
COMPARISON OF FUNCTIONALITY

Scheme	Hetero.	SPF	Attack Type		Infra
			IA	MMA	
SL [50]	×	✓	✓	✓	SMPC
PPFL [51]	×	✓	✓	✓	DP
FBI [52]	×	✓	✓	✓	DP
DHFL [53]	×	✓	✓	✓	HE
PROPOSED	✓	✓	✓	✓	PKI-IB-HAAGKA

TABLE III
CNN ARCHITECTURES ARGUMENT FOR EXPERIMENT

Definition	Value
Fraction of clients (C)	0.1
Number of local epochs (E)	5
Local batch size (B)	10
Test batch size (B)	128
Learning rate	0.01
SGD momentum	0.5
Number of each kind of kernel	9
Kernel size to use for convolution	3,4,5
Number of filters for conv nets	32

VI. EVALUATION AND ANALYSIS

The evaluation, security analysis, and performance analysis of the proposed HPP-BEFL will be given in this section.

A. Functionality Evaluation

This article compared the functionality of four state-of-art existing schemes: SL [50], PPFL [51], FBI [52], and DHFL [53] with the proposed HPP-BEFL from four perspectives.

1) *Crypto System Heterogeneity (Hetero.)*: whether system could supports heterogeneous crypto system.
2) *Single Point of Failure (SPF)*: whether system could solve the SPF problem.
3) *Attack Type*: whether system could resist inference attack (IA) and the MMA.
4) *Infrastructure (Infra)*: underlining technology.

As illustrated in Table II, all four existing schemes and the proposed scheme could resist the SPF for deploying the federated learning system in the blockchain environment. As a consequence of the implementation of different privacy-preserving technologies, including secure multiparty computation (SMPC), differential privacy (DP), and homomorphic encryption (HE), all compared schemes can successfully protect against IA and MMA. However, none of the existing schemes support blockchain system server nodes from heterogeneous crypto system i.e., support the heterogeneity of crypto system. Moreover, it is noteworthy that the proposed scheme is the first to introduce the utility of a heterogeneous asymmetric group key agreement scheme as an infrastructure for achieving privacy-preserving.

B. Security Analysis

The security analysis of the proposed scheme will be discussed here.

Theorem 3: Faced with the IA and the MMA, the proposed HPP-BEFL's privacy-preserving can be deduced from the confidentiality of the proposed PKI-IB-HAAGKA.

Proof: The proposed HPP-BEFL is based on the building block PKI-IB-AAGKA, which has been proven to be secure against IND-CPA under the k-BDHE assumption. With the confidentiality of the PKI-IB-HAAGKA, the encrypted local trained model is indistinguishable. Thus, the attackers could not execute MMA and IA directly to steal the privacy data, i.e., guaranteeing the privacy-preserving of the proposed HPP-BEFL.

C. Performance Analysis

Based on experiments, the performance analysis of the proposed scheme will be discussed here.

1) Experimental Settings:

a) *Environment*: All schemes run on the same experiment environment (Linux Ubuntu 20.04, Intel Xeon(R) CPU E52650 v4 @ 2.20GHz, Tesla P40 GPU, 32G RAM, 2 × 2T HDD).
b) *Dataset*: All schemes conducted experiments with two standard image datasets: CIFAR-10 [10] and MNIST [9]. The MNIST dataset consists of 70 000 grayscale images of handwritten digits, categorized into 10 distinct classes. Prior to being input into the convolutional neural network (CNN), each image is normalized to a standard size of 28×28 pixels. In comparison, the CIFAR-10 dataset comprises 60 000 color images, each with dimensions of 32×32 pixels, organized into 10 categories, including cars, dogs, and horses, with each category containing 6,000 images.
c) *Model*: the proposed HPP-BEFL implemented using a CNN architecture is listed in Table III which is well-suited for training on both the MNIST and CIFAR-10 datasets.
d) *Dependencies*: the underlying PKI-IB-HAAGKA protocol was implemented in C++ using the PBC library [54]. The proposed HPP-BEFL system baseline was implemented in Python with PyTorch [55], OpenCV [56], and numpy [57].

2) Training Accuracy and Training Loss: Multiple experiments are conducted to evaluate the training loss and training accuracy of the proposed HPP-BEFL with different participants size ($N = 20, 40, 60, 80, 100$).

Fig. 3 illustrates the changes in training loss and training accuracy of the proposed HPP-BEFL as the number of participants varies with increasing aggregation rounds. It can be observed that regardless of the number of participants, the global model converges after a certain number of aggregation rounds. As the number of local computing devices in a federated learning system increases, the convergence rate of the global model tends to decelerate, often accompanied by a reduction in accuracy. This trend arises from the partitioning of a fixed-size dataset among a larger pool of participants, which decreases the quantity of training data available at each node, thereby hindering the model's ability to capture the underlying

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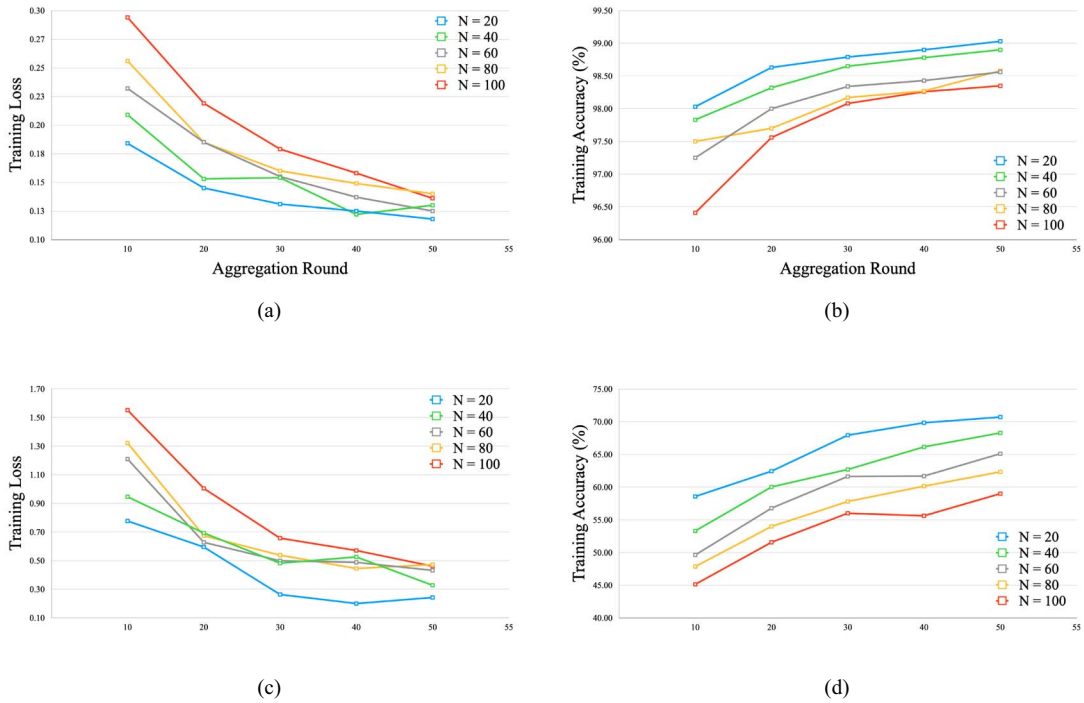


Fig. 3. Training loss/accuracy on MNIST/CIFAR-10. (a) MNIST/training loss. (b) MNIST/training accuracy. (c) CIFAR-10/training loss. (d) CIFAR-10/training accuracy.

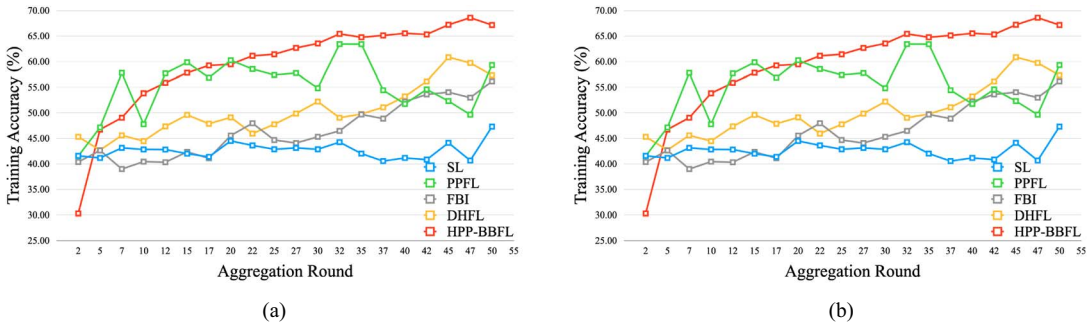


Fig. 4. Comparison of training accuracy on MNIST/CIFAR-10. (a) MNIST/training accuracy. (b) CIFAR-10/training accuracy.

features and patterns of the data. For instance, even with $N = 20$, the MNIST dataset achieves an accuracy of 99.03% after 50 aggregation rounds, while the CIFAR-10 dataset demonstrates a comparatively lower accuracy of 70.71%. This disparity is primarily attributed to the increased complexity and variability of CIFAR-10 relative to MNIST; the three-channel color images in CIFAR-10 present greater challenges for convergence, such that a simpler model architecture or insufficient training duration may impede the model's ability to achieve full convergence.

As shown in Fig. 4, this article compares the proposed HPP-BEFL scheme with four previously discussed schemes [50], [51], [52], [53] regarding training accuracy, with 40 participants using the MNIST and CIFAR-10 datasets. It is observed that after several training rounds, the training accuracy of all five schemes converge to certain values. Moreover, since this scheme does not introduce noise into the model such as PPFL and FBI, its training accuracy is significantly higher, achieving 68.6% (CIFAR-10) and 98.88% (MNIST). Thus, it can be concluded that the accuracy of the model trained by the HPP-BEFL scheme is not compromised by the introduction of the new privacy-preserving mechanism PKI-IB-HAAGKA.

3) Efficiency: Regarding the efficiency of the proposed PKI-IB-HAAGKA scheme, this article compares it with the cryptographic solutions used in SL [50] and DHFL [53], namely SMPC and HE, focusing on encryption time, decryption time, and total time for encryption and decryption. As illustrated in Fig. 5, the average time for both encryption and decryption of PKI-IB-HAAGKA is significantly lower than that of the other schemes, indicating that this solution has exceptionally high encryption and decryption efficiency. Furthermore, it is noteworthy that the key agreement time of PKI-IB-HAAGKA in BEFL is only related to the number of aggregation nodes in the blockchain and is independent of the number of local

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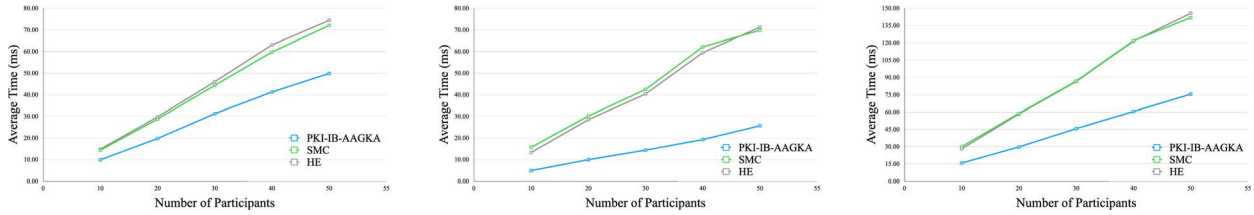


Fig. 5. Comparison of encryption and decryption time. (a) Enc time. (b) Dec time. (c) Enc + Dec time.

computing devices. Therefore, the HPP-BEFL proposed in this article is an efficient federated learning scheme suitable for large-scale training.

What's more, the proposed PKI-IB-HBMS and PKI-IB-HAAGKA protocol achieves linear communication complexity. It's observed that only the Agreement and Sign/Encrypt phases incur communication costs. Theoretical calculations indicate that the communication cost of Agreement increases linearly with the group size, with a theoretical value of $(n+1)L_1$, where n is the group size and L_1 is the length of an element in $\mathbb{G}_1$. The ciphertext size associated with Sign/Encrypt reflects a constant communication cost, with a theoretical value of $4L_1 + L_m$, where L_m is the length of an element in plaintext.

VII. Conclusion and Future Works

To address current problems social fintech facing with, the article introduces a novel PKI-IB-HAAGKA, which effectively resolves issues related to cryptographic system heterogeneity. Additionally, a PKI-IB-HBMS scheme is proposed as a building block of PKI-IB-HAAGKA. Building on this, the article presents the HPP-BEFL system, which enhances privacy, security, and efficiency in social fintech applications by mitigating inference attacks and improving system efficiency. Through security analysis and experimental results, the article demonstrates that the proposed PKI-IB-HAAGKA, PKI-IB-HBMS, and HPP-BEFL are provably secure, efficient against security attacks, and highly efficient, making them suitable for large-scale heterogeneous privacy-preserving model training scenarios.

However, the proposed HPP-BEFL primarily addresses the security of the local model upload process, overlooking the security aspects of the global model download. Therefore, it is essential to explore additional approaches to secure the downloading process. Furthermore, while the scheme focuses on the heterogeneity of PKI and identity-based cryptographic systems, it does not account for other heterogeneous cryptographic systems and various forms of heterogeneity present in real-world applications. Consequently, identifying and developing solutions for these additional heterogeneous scenarios represents a significant avenue for future research.

Appendix A
Notations of Building Block

Here, for convenience, symbols that appeared in the building block are listed in Table IV.

TABLE IV
Symbols in Building Block

Symbols	Meaning
n	Number all protocol participants
$n_{\mathrm{PKI}/ID}$	Number of PKI or identity-based participants
i	Index of system participants
ID_i	Identity of $\mathcal{U}_{ID_i}$
id_i	Hash value of ID_i
$\mathcal{U}_{\mathrm{PKI}/ID_i}$	Participant i from correspoding system
κ	Master secret
pk_{PKI/ID_i}	Public key of $\mathcal{U}_i$
sk_{PKI/ID_i}	Private key of $\mathcal{U}_i$
$\sigma_{\mathrm{PKI}/ID_i}$	Batch signature slice of $\mathcal{U}_i$
ρ_{PKI/ID_i}	PKI or identity-based signature of $\mathcal{U}_i$
cert_i	Certificate of $\mathcal{U}_{\mathrm{PKI}i}$ from CA
s_i	Secret key of $\mathcal{U}_{ID_i}$
q	Big prime number
$\mathbb{Z}_q$	Finite field
$\mathbb{G}_1, \mathbb{G}_2$	Multiplicative cyclic group
g	Generator of $\mathbb{G}_1$
$\hat{e}$	Bilinear mapping
H_1, H_2, H_3	Hash functions
ek	Encode key of the whole group
dk_i	Decryption key of $\mathcal{U}_i$
m_i	Messages to be signed
$\mathbb{M}_{\mathrm{PKI}/ID_i}$	Published message of $\mathcal{U}_i$
f_i	Hash value of m_i or i
m	Plain text
ζ	Bit-size of the plaintext
c	Cipher text
$\mathrm{pid}^{\chi}_{\mathcal{U}_i}$	Set of all participant's index
$\mathrm{sid}^{\chi}_{\mathcal{U}_i}$	Session ID of $\Omega^{\chi}_{\mathcal{U}}$
$ms^{\chi}_{\mathcal{U}_i}$	All messages generated by $\Omega^{\chi}_{\mathcal{U}}$
$ek^{\chi}_{\mathcal{U}_i}$	Encryption key held by $\Omega^{\chi}_{\mathcal{U}}$
$dk^{\chi}_{\mathcal{U}_i}$	Decryption key held by $\Omega^{\chi}_{\mathcal{U}}$
$\mathrm{state}^{\chi}_{\mathcal{U}_i}$	Internal state of instance $\Omega^{\chi}_{\mathcal{U}}$
$h_i, \eta_{\mathrm{PKI}/ID_i}, e$	Random value of $\mathcal{U}_i$
$r_{\mathrm{PKI}/ID_i}, d_{\mathrm{PKI}/ID_i}$	Agreement slices of $\mathcal{U}_i$
Q, ω, d_j	Aggregated item

Appendix B
Definition of Building Block

Definition 7: Partnering.

Suppose instances $\Omega_{\mathcal{U}_i}$ and $\Omega_{\mathcal{U}_j}$ where $i \neq j$ are partnered if

1) They are *successfully terminated*;
2) $\mathrm{pid}^{\chi}_{\mathcal{U}_i} = \mathrm{pid}^{\chi'}_{\mathcal{U}_j}$
3) $\mathrm{sid}^{\chi}_{\mathcal{U}_i} = \mathrm{sid}^{\chi'}_{\mathcal{U}_j}$

Definition 8:

Freshness. The freshness of the instance could be defined as follows.

1) $\Omega^{\chi}_{\mathcal{U}_i}$ has never been queried with $\mathrm{Dk.Reveal}(\Omega^{\chi}_{\mathcal{U}_i})$ or $\mathrm{Dk.Reveal}(\Omega^{\chi'}_{\mathcal{U}_i})$ by $\mathcal{A}$, where $\Omega^{\chi}_{\mathcal{U}_i}$ is partnered with $\Omega^{\chi'}_{\mathcal{U}_i}$.

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2) PKI-Corrupt($\Omega^{\chi}_{\mathcal{U}_{\mathrm{PKI}_i}}$) and ID-Corrupt($\Omega^{\chi}_{\mathcal{U}_{ID_i}}$) have never been queried before PKI-ID-Send($\Omega^{\chi}_{\mathcal{U}_i}, \Phi$).
3) At least one private keys of $\mathrm{sid}^{\chi}_{\mathcal{U}_{\mathrm{PKI}/ID}}$ are not corrupted.

Definition 9: Terminated.

$\mathrm{state}^{\chi}_{\mathcal{U}_i}$ will be labeled with terminated when all message sending and receiving have finished. The termination (accept) of PKI-IB-AAGKA protocol will be flagged by the instance $\Omega^{\chi}_{\mathcal{U}}$ with $(ek^{\chi}_{\mathcal{U}_i}, dk^{\chi}_{\mathcal{U}_i}, \mathrm{pid}^{\chi}_{\mathcal{U}_i}, \mathrm{sid}^{\chi}_{\mathcal{U}_i})$ where $ek^{\chi}_{\mathcal{U}_i} \neq \mathrm{null}$ and $dk^{\chi}_{\mathcal{U}_i} \neq \mathrm{null}$.

Appendix C
Security Model of PKI-IB-HBMS

The security of the proposed PKI-IB-HBMS is illustrated by the game between an adversary $\mathcal{A}$ and a challenger $\mathcal{C}$ as follows.

1) *Initial*: First, $\mathcal{C}$ could collect the public parameter list pp and the master-secret κ by executing HBMS.Setup. Then, $\mathcal{C}$ sends pp to $\mathcal{A}$ but preserving κ.
2) *Training*: $\mathcal{A}$ could make the following types of queries to $\mathcal{C}$ adaptively,with a polynomially bounded number of times.
   a) *HBMS.PKI-Extract Query*: The public–private key pair of a participant from PKI crypto system could be queried by $\mathcal{A}$ from $\mathcal{C}$ with the index i. On receiving such a query, $\mathcal{C}$ responds with $(pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i})$
   b) *HBMS.ID-Extract Query*: The public–private key pair of a participant from identity-based crypto system could by queried by $\mathcal{A}$ from $\mathcal{C}$ with the identity ID_i. On receiving such a query, $\mathcal{C}$ responds with the corresponding (pk_{ID_i}, sk_{ID_i})
   c) *PKI-Sign Query*: The batch signature on t message of a participant from PKI crypto system could be queried by $\mathcal{A}$ from $\mathcal{C}$ with the index ID_i. On receiving such a query, $\mathcal{C}$ responds with $\sigma_{\mathrm{PKI}i}$
   d) *ID-Sign Query*: The batch signature on t message of a participant from identity-based crypto system could be queried by $\mathcal{A}$ from $\mathcal{C}$ with the identity ID_i. On receiving such a query, $\mathcal{C}$ responds with σ_{ID_i}
3) *Forgery*: $\mathcal{A}$ produces $\{\mathbb{L}^*, \sigma^*, m^*\}$ where $\mathbb{L}^* = \{pk_{\mathrm{PKI}/ID_1^*}, \ldots, pk_{\mathrm{PKI}/ID_n^*}\}$ is a public key set, along with a multisignature σ^* and a message m^*. It's said that $\mathcal{A}$ wins the above game if the following conditions are met.
   a) σ^* is valid multisignature on message m^* under public key set $\{pk_{\mathrm{PKI}/ID_i^*}\}_{1\leq i\leq n}$.
   b) At least one related index i or identity ID_i of public key in the set $\mathbb{L}^*$ has never been queried by $\mathcal{A}$ with HBMS.PKI.Extract or the HBMS.ID.Extract. What's more, $\mathcal{A}$ has never executed the HBMS.PKI-Sign and HBMS.ID-Sign with m^* together with the public key.

Appendix D
Proof of Theorem 1

Let $\mathcal{C}$ be a challenger and $\mathcal{A}$ an adversary capable of breaking the proposed PKI-IB-HBMS with adaptive chosen-message attack. Assume that $\mathcal{C}$ given $(g, g^{\alpha}, g^{\beta})$ in $\mathbb{G}_1$, which represents an instance of the CDH problem. The following part illustrate how $\mathcal{C}$ could utilize $\mathcal{A}$ to address the CDH problem, i.e. to compute $g^{\alpha\beta}$, with advantage.

1) *Initial*: First, $\mathcal{C}$ sets $g_1 = g^{\alpha}$, then chooses $pp = (\mathbb{G}_1, \mathbb{G}_2, \hat{e}, H_1, H_2, g, g_1)$ as system public parameters, and send pp to $\mathcal{A}$. In what follows, $\mathcal{C}$ controls H_1 and H_2 considered as random oracles.
2) *Training*: On receiving $\mathcal{A}$'s queries, $\mathcal{C}$ responds as follows.
   a) H_1 *Query*: There is an initially empty list $L_1 = (ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \mathrm{coin}_i^{L_1})$ kept by $\mathcal{C}$. On receiving queries with identity ID_i from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
      – If ID_i is founded in L_1, responds with corresponding id_i directly.
      – Else flips $\mathrm{coin}_i^{L_1}$ from $\{0,1\}$, picks $\mu_i \in Z_q^*$ randomly and performs as follows.
         * If $\mathrm{coin}_i^{L_1} = 0$ with probability p, computes $id_i = g^{\mu_i}, pk_{ID_i} = \hat{e}(id_i, g_1), sk_{ID_i} = g_1^{\mu_i}$, adds $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \mathrm{coin}_i^{L_1})$ to L_1 and responds with id_i.
         * Else with probability $1-p$ computes $id_i = g^{\beta\mu_i}, pk_{ID_i} = \hat{e}(id_i, g_1), sk_{ID_i} = \mathrm{null}$, adds $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \mathrm{coin}_i^{L_1})$ to L_1 and responds with id_i.
   b) H_2 *Query*: There is an initially empty list $L_2 = (m_i, f_i, v_i, \mathrm{coin}_i^{L_2})$ kept by $\mathcal{C}$. On receiving queries with message m_i from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
      – If m_i is founded in L_2, responds with corresponding f_i directly.
      – Else flips $\mathrm{coin}_i^{L_2}$ from $\{0,1\}$, picks $v_i \in Z_q^*$ randomly and performs as follows.
         * If $\mathrm{coin}_i^{L_2} = 0$ with probability p, computes $f_i = g^{v_i} g^{\alpha}$ and adds $(m_i, f_i, v_i, \mathrm{coin}_i^{L_2})$ to L_2 and responds with f_i.
         * Else with probability $1-p$ computes $f_i = g^{v_i}$ adds $(m_i, f_i, v_i, \mathrm{coin}_i^{L_2})$ to L_2 and responds with f_i.
   c) *PKI-Extract Query*: There is an initially empty list $L_{\mathrm{PKI}} = (i, h_i, pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i}, o_i, \mathrm{coin}_i^{L_{\mathrm{PKI}}})$ kept by $\mathcal{C}$. On receiving queries with index i of $\mathcal{U}_{\mathrm{PKI}i}$ from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
      – If i is founded in L_{PKI}, responds with corresponding $(pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i})$ directly.
      – Else flips $\mathrm{coin}_i^{L_{\mathrm{PKI}}}$ from $\{0,1\}$, picks $o_i \in Z_q^*$ randomly and performs as follows.
         * If $\mathrm{coin}_i^{L_{\mathrm{PKI}}} = 0$ with probability p, computes $sk_{\mathrm{PKI}i} = h_i = g^{o_i}, pk_{\mathrm{PKI}i} = \hat{e}(sk_{\mathrm{PKI}i}, g)$, adds $(i, h_i, pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i}, o_i, \mathrm{coin}_i^{L_{\mathrm{PKI}}})$ to L_{PKI} and responds with $(pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i})$.
         * Else with probability $1-p$ computes $h_i = g^{\beta\mu_i}$, $pk_{\mathrm{PKI}i} = \hat{e}(h_i, g), sk_{\mathrm{PKI}i} = \mathrm{null}$, adds $(i, h_i, pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i}, o_i, \mathrm{coin}_i^{L_{\mathrm{PKI}}})$ to L_{PKI} and responds with $(pk_{\mathrm{PKI}i}, sk_{\mathrm{PKI}i})$.

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d) *ID-Extract Query*: On receiving queries with identity ID_i from $\mathcal{U}_{ID_i}$, $\mathcal{C}$ executes H_1 query with ID_i and then recover the corresponding item $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{L_1})$ from L_1 and responds with (pk_{ID_i}, sk_{ID_i}).

e) *PKI-IB-Sign query*:

i) On receiving queries with $(i, m_1 \dots m_t)$ $t \leq N$, $\mathcal{C}$ executes the PKI-extract query with index i and recovers $(i, h_i, pk_{\text{PKI}i}, sk_{\text{PKI}i}, o_i, \text{coin}_i^{L_{\text{PKI}}})$ from L_{PKI}. Then, $\mathcal{C}$ executes the H_2 query with $\{m_1 \dots m_t\}$ with $t \leq N$ then recovers $(m_i, f_i, v_i, \text{coin}_i^{L_2})$ from L_2 and performs as follows.
* If $\text{coin}_i^{L_{\text{PKI}}} = 0$, responds with PKI-IB-Sign algorithm directly.
* Else if $\text{coin}_i^{L_2} = 0$, selects $\eta_{\text{PKI}i} \in \mathbb{Z}_q^\star$ randomly, computes $r_{\text{PKI}i} = g^{\eta_{\text{PKI}i}} g^{-\beta o_i}$, $d_{i,j} = r_{\text{PKI}i}^{v_j} g_{(1\leq j\leq t)}^{\eta_{\text{PKI}i}}$ and responds with $(r_{\text{PKI}i}, \{d_{\text{PKI}i,j}\}_{1\leq j\leq t})$.
* Else Abort.

ii) On receiving queries with $(ID_i, m_1, \dots, m_t)$ with $t \leq N$, $\mathcal{C}$ executes the H_1 query with identity ID_i and recovers $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{L_1})$ from L_1. Then, $\mathcal{C}$ executes the H_2 query with $\{m_1 \dots m_t\}$ then recovers $(m_i, f_i, v_i, \text{coin}_i^{L_2})$ from L_2 and performs as follows.
* If $\text{coin}_i^{L_{\text{PKI}}} = 0$, responds with PKI-IB-Sign algorithm directly.
* Else if $\text{coin}_i^{L_2} = 0$, selects $\eta_{ID_i} \in \mathbb{Z}_q^\star$ randomly, computes $r_{ID_i} = g^{\eta_{ID_i}} g^{-\beta\mu_i}$, $d_{i,j} = r_{ID_i}^{v_j} g_1^{\eta_{ID_i}}$ and responds with $(r_{ID_i}, \{d_{ID_i,j}\}_{1\leq j\leq t})$.
* Else abort.

3) *Forgery*: Eventually, $\mathcal{A}$ produces $\{\mathbb{L}^*, \sigma^*, m^*\}$, where $\mathbb{L}^* = \{pk_{\text{PKI}/ID_1}^*, \dots, pk_{\text{PKI}/ID_n}^*\}$ is a public key set, along with a valid multisignature σ^* on message m^*.

At this point, $\mathcal{C}$ could solve the CDH problem as follows.

a) For all $pk_{ID_i}^*$ in $\mathbb{L}^*$, executes H_1 query with the corresponding ID_i^*, and recovers $(ID_i^*, id_i^*, pk_{ID_i}^*, sk_{ID_i}^*, \mu_i^*, \text{coin}_i^{L_1{}^*})$ from L_1.

b) Execute H_2 query ,with m^*, and recovers $(m_i^*, f_i^*, v_i, \text{coin}_i^{L_2{}^*})$ from L_2.

c) For all $pk_{\text{PKI}i}^*$ in $\mathbb{L}^*$, executes PKI-extract query with the corresponding index i, and recovers $(i^*, h_i^*, pk_{\text{PKI}i}^*, sk_{\text{PKI}i}^*, o_i^*, \text{coin}_i^{L_{\text{PKI}}{}^*})$ from L_{PKI}.

In the subsequent discussion, for clarity, this article will concentrate on the case that $\text{coin}_1^{L_1{}^*} = \text{coin}^{L_2{}^*} = 1$, $\text{coin}^{L_{\text{PKI}}{}^*} = 0$ and $\text{coin}_{i\ (2\leq i\leq t)}^{L_1{}^*} = 0$; If not, aborts.

If the process continues, $\mathcal{C}$ could computes $sk_{\text{PKI}}{}^* = g^{o^\star}$, $pk_{\text{PKI}}{}^* = \hat{e}(sk_{\text{PKI}}{}^*, g)$, $sk_{ID_1}^* = g^{\beta\mu_1^*}$, $pk_{ID_1}^* = \hat{e}(id_1^*, g_1)$, $\{sk_{ID_i}^* = g_1^{\mu_i^*}, pk_{ID_i}^* = \hat{e}(id_i^*, g_1)\}_{2\leq i\leq t}$, $f^* = g^{v^*}$. Because σ^* is a valid multisignature, $\mathcal{C}$ could computes $\hat{e}(d^*, g) = \hat{e}(f^*, \omega^*) \cdot Q^*$. From the above equation it's easy for $\mathcal{C}$ to compute the solution of the CDH problem as $g^{\alpha\beta} = (g_1^{-\sum_{i=2}^{n_{ID}} \mu_i^*} g^{-o^*} \cdot d^* \cdot \omega^{*-v^*})^{\mu_1^{*-1}}$.

APPENDIX E
SECURITY MODEL OF PKI-IB-HAAGKA

The security of the proposed PKI-IB-HAAGKA is illustrated by the game between an adversary $\mathcal{A}$ and a challenger $\mathcal{C}$ as follows.

1) *Initial*: First, $\mathcal{C}$ could collect the public parameter list pp and the master-secret κ by executing Setup. Then, $\mathcal{C}$ sends pp to $\mathcal{A}$ but preserving κ.

2) *Training*: $\mathcal{A}$ could make the following types of queries to $\mathcal{C}$ adaptively, with a polynomially bounded number of times.

a) *PKI-IB-Send* $(\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}}, \Phi)$: On receiving the message Φ to $\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}}$, responds with what produces by $\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}}$.

b) *PKI-Corrupt* $(\Omega^\chi_{\mathcal{U}_{\text{PKI}i}})$: On receiving the index i of $\Omega^\chi_{\mathcal{U}_{\text{PKI}i}}$, responds with the private key of $\Omega^\chi_{\mathcal{U}_{PKIi}}$.

c) *ID-Corrupt* $(\Omega^\chi_{\mathcal{U}_{ID_i}})$: On receiving the identity ID_i of $\Omega^\chi_{\mathcal{U}_{ID_i}}$, responds with the private key of $\Omega^\chi_{\mathcal{U}_{ID_i}}$.

d) *Enc.Reveal* $(\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}})$: On receiving the identity ID_i of $\Omega^\chi_{\mathcal{U}_{ID_i}}$ or the index i of $\Omega^\chi_{\mathcal{U}_{\text{PKI}i}}$, responds with the corresponding encryption key $\text{ek}_{\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}}}$.

e) *Dec.Reveal* $(\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}})$: On receiving the identity ID_i of $\Omega^\chi_{\mathcal{U}_{ID_i}}$ or the index i of $\Omega^\chi_{\mathcal{U}_{\text{PKI}i}}$, responds with the corresponding decryption key $\text{dk}_{\Omega^\chi_{\mathcal{U}_{\text{PKI}/ID_i}}}$.

3) *Test* $(\Omega^\chi_{\mathcal{U}_i})$: Eventually, $\mathcal{A}$ produces $\Omega^\chi_{\mathcal{U}_i}, (m_0, m_1)$ where $|m_0| = |m_1|$ and $\Omega^\chi_{\mathcal{U}_i}$ should be fresh (See Definition B). And then, $\mathcal{C}$ selects $b \in \{0, 1\}$ randomly, encrypts m_b with $\text{ek}_{\Omega^\chi_{\mathcal{U}_i}}$ resulting in c, and sends c to $\mathcal{A}$.

4) *Response*: $\mathcal{A}$ produces b'. If $b' = b$, $\mathcal{A}$ wins with the advantage $\text{Adv}(\mathcal{A}) = |\, 2\Pr[b = b'] - 1\,|$.

APPENDIX F
PROOF OF THEOREM 2

Let $\mathcal{C}$ be a challenger and $\mathcal{A}$ an adversary capable of breaking the proposed PKI-IB-AAGKA. Assume that $\mathcal{C}$ given the instance of the k-BDHE problem $(g, h, \{y_i\}_{i\in\{1,\dots,k,k+2,\dots,2k\}})$, where $y_i = g^{\alpha^i}$. The following part illustrate how $\mathcal{C}$ could utilize $\mathcal{A}$ to address the k-BDHE problem, i.e. to compute $\hat{e}(g, h)^{\alpha_{k+1}}$, with advantage.

Hereafter, the group size in each session is limited to k. Each new session is bounded with a unique session ID sid_ι. $\mathcal{C}$ will flip a coin $\text{coin}_{\text{sid}_\iota}$ and record $(\text{sid}_i, \text{coin}_{\text{sid}_\iota})$.

1) *Initial*: First, $\mathcal{C}$ sets $g_1 = y_1 = g^\alpha$, then picks $pp = (\mathbb{G}_1, \mathbb{G}_2, \hat{e}, g, g_1, \varsigma, H_1, H_2, H_3)$ as the system public parameters, and send pp to $\mathcal{A}$. In what follows, $\mathcal{C}$ controls H_1, H_2, and H_3 viewed as random oracles.

2) *Training*: On receiving $\mathcal{A}$'s queries, $\mathcal{C}$ responds as follows.

a) H_1 *Queries*: There is an initially empty list $\mathcal{L}_1 = (ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{\mathcal{L}_1})$ maintained by $\mathcal{C}$. On receiving queries with identity ID_i from $\mathcal{A}$, $\mathcal{C}$ performs as follows.

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- If ID_i could be founded in $\mathcal{L}_1$, responds with corresponding id_i directly.
- Else flips $\text{coin}_i^{\mathcal{L}_1}$ from $\{0,1\}$, picks $\mu_i \in Z_q^*$ randomly and performs as follows
  * If $\text{coin}_i^{L_1} = 0$ with probability p, computes $id_i = g^{\mu_i}, pk_{ID_i} = \hat{e}(id_i, g_1), sk_{ID_i} = g_1^{\mu_i}$, adds $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{\mathcal{L}_1})$ to $\mathcal{L}_1$ and responds with id_i.
  * Else with probability $1-p$ computes $id_i = g^{\mu_i} y_k, pk_{ID_i} = \hat{e}(id_i, g_1), sk_{IDi} = \text{null}$, adds $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{\mathcal{L}_1})$ to $\mathcal{L}_1$ and responds with id_i.

b) H_2 *Query*: There is an initially empty list $\mathcal{L}_2 = (i, f_i, v_i)$ kept by $\mathcal{C}$. On receiving queries with message m_i from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
- If m_i is founded in $\mathcal{L}_2$, responds with corresponding f_i directly.
- Else picks $v_i \in Z_q^*$ randomly and performs as follows.
  * If $i < k$, computes $f_i = g^{v_i} y_i$ and adds (i, f_i, v_i) to $\mathcal{L}_2$ and responds with f_i.
  * Else computes $f_i = g^{v_i}$ adds (i, f_i, v_i) to L_2 and responds with f_i.

c) H_3 *Queries*: There is an initially empty list $\mathcal{L}_3 = (\Phi_i, \Psi_i)$ kept by $\mathcal{C}$. On receiving queries with message Φ_i from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
- If Φ_i could be found in H_3^{list}, responds with Ψ_i directly.
- Else randomly chooses $\Psi_i \in \{0,1\}^{\varsigma}$, adds (Φ_i, Ψ_i) to H_3^{list} and responds with Φ_i.

d) *PKI-Corrupt*($\mathcal{U}_{\text{PKI}i}$): There is an initially empty list $\mathcal{L}_{\text{PKI}} = (i, h_i, pk_{\text{PKI}i}, sk_{\text{PKI}i}, o_i, \text{coin}_i^{\mathcal{L}_{\text{PKI}}})$ kept by $\mathcal{C}$. On receiving queries with index i of $\mathcal{U}_{\text{PKI}i}$ from $\mathcal{A}$, $\mathcal{C}$ performs as follows.
- If i is founded in $\mathcal{L}_{\text{PKI}}$, responds with corresponding $(pk_{\text{PKI}i}, sk_{\text{PKI}i})$ directly.
- Else flips $\text{coin}_i^{\mathcal{L}_{\text{PKI}}}$ from $\{0,1\}$, picks $o_i \in Z_q^*$ randomly and performs as follows.
  * If $\text{coin}_i^{\mathcal{L}_{\text{PKI}}} = 0$ with probability p, computes $sk_{\text{PKI}i} = h_i = g^{o_i}, pk_{\text{PKI}i} = \hat{e}(sk_{\text{PKI}i}, g)$, adds $(i, h_i, pk_{\text{PKI}i}, sk_{\text{PKI}i}, o_i, \text{coin}_i^{\mathcal{L}_{\text{PKI}}})$ to $\mathcal{L}_{\text{PKI}}$ and responds with $(pk_{\text{PKI}i}, sk_{\text{PKI}_i})$.
  * Else with probability $1-p$ computes $h_i = g^{\mu_i} y_k, pk_{\text{PKI}i} = \hat{e}(h_i, g), sk_{\text{PKI}i} = \text{null}$, adds $(i, h_i, pk_{\text{PKI}i}, sk_{\text{PKI}i}, o_i, \text{coin}_i^{\mathcal{L}_{\text{PKI}}})$ to $\mathcal{L}_{\text{PKI}}$ and responds with $(pk_{\text{PKI}i}, sk_{\text{PKI}_i})$.

e) *ID-Corrupt*($\mathcal{U}_{ID_i}$): On receiving queries with identity ID_i from $\mathcal{U}_{ID_i}$, $\mathcal{C}$ executes H_1 query with ID_i and then recovers the corresponding item $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{\mathcal{L}_1})$ from $\mathcal{L}_1$, and performs as follows.
- If $\text{coin}_i^{\mathcal{L}_1} = 0$, responds with (pk_{ID_i}, sk_{ID_i}).
- Else abort

f) *PKI-IB-Send*($\Omega_{\mathcal{U}_i}^{\chi}, \Phi$): There are two initially empty lists $\mathcal{S}_{\text{PKI}} = (i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{\text{PKI}i}, d_{\text{PKI}i,i}), \mathcal{S}_{ID} = (ID_i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{ID_i}, d_{ID_i,i})$. On receiving queries with the message Φ, $\mathcal{C}$ recovers $(\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}}, \text{sid}_{\mathcal{U}_i}^{\chi})$, $(ID_i, id_i, pk_{ID_i}, sk_{ID_i}, \mu_i, \text{coin}_i^{\mathcal{L}_1})$, (m_j, f_j, v_j), $(i, h_i, pk_{\text{PKI}i}, sk_{\text{PKI}i}, o_i, \text{coin}_i^{\mathcal{L}_{\text{PKI}}})$ by executing H_1 query, H_2 query, PKI-corrupt query and ID-corrupt query if any of the queries has never been executed before. And then $\mathcal{C}$ performs as follows.

i) On input the index i of $\mathcal{U}_{\text{PKI}i}$
- If $\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}} = 0$ and $\text{coin}_i^{\mathcal{L}_{\text{PKI}}} = 0$
  a) Picks $\eta_{\text{PKI}i} \in \mathbb{Z}_q^*$ randomly and computes: $r_{\text{PKI}i} = g^{\eta_{\text{PKI}i}}$
  b) Computes $\{d_{\text{PKI}i,j}\}_{1 \le j \le n}$ where $d_{\text{PKI}i,j} = sk_{\text{PKI}i} f_j^{\eta_{\text{PKI}i}}$
  c) Adds $(i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{\text{PKI}i}, d_{\text{PKI}i,i})$ to $\mathcal{S}_{\text{PKI}}$ and responds with $(r_{\text{PKI}i}, \rho_{\text{PKI}i}, \{d_{\text{PKI}i,j}\}_{1 \le j \le n, j \ne i})$
- Else if $\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}} = 0$ and $\text{coin}_i^{\mathcal{L}_{\text{PKI}}} = 1$
  a) Picks $\eta_{\text{PKI}i} \in \mathbb{Z}_q^*$ randomly and computes $r_{\text{PKI}i} = g^{\eta_{\text{PKI}i}} \prod_{l=1}^{n} y_{k-l+1}^{-1}$,
  b) Computes $\{d_{\text{PKI}i,j}\}_{1 \le j \le n}$ where $d_{\text{PKI}i,j} = {r_{\text{PKI}i}}^{v_j} g^{o_i} y_j^{\eta_{\text{PKI}i}} \prod_{l=1}^{n, l \ne j} y_{k-l+1+j}^{-1}$
  c) Adds $(i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{\text{PKI}i}, d_{\text{PKI}i,i})$ to $\mathcal{S}_{\text{PKI}}$ and responds with $(r_{\text{PKI}i}, \rho_{\text{PKI}i}, \{d_{\text{PKI}i,j}\}_{1 \le j \le n, j \ne i})$
- Else if $\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}} = 1$ and $\text{coin}_i^{\mathcal{L}_{\text{PKI}}} = 0$
  a) Picks $\eta_{\text{PKI}i} \in \mathbb{Z}_q^*$ randomly and computes $r_{\text{PKI}i} = g^{\eta_{\text{PKI}i}} y_{k-i+1}$.
  b) Computes $\{d_{\text{PKI}i,j}\}_{1 \le j \le n}$ where $d_{\text{PKI}i,j} = {r_{\text{PKI}i}}^{\nu_j} g^{o_i} y_j^{\eta_{\text{PKI}i}} y_{k-i+1+j}$
- Else, $\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}} = 1$ and $\text{coin}_i^{\mathcal{L}_{\text{PKI}}} = 1$
  a) Picks $\eta_{\text{PKI}i} \in \mathbb{Z}_q^*$ randomly and computes: $r_{\text{PKI}i} = g^{\eta_{\text{PKI}i}} \prod_{l=1}^{n, l \ne i} y_{k-l+1}^{-1}$
  b) Computes $\{d_{\text{PKI}i,j}\}_{1 \le j \le n}$ where $d_{\text{PKI}i,j} = {r_{\text{PKI}i}}^{\nu_j} g^{o_i} y_j^{\eta_{\text{PKI}i}} \prod_{l=1}^{n, l \notin \{i,j\}} y_{k-l+1+j}^{-1}$
  c) Adds the item $(i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{\text{PKI}i}, \text{null})$ to $\mathcal{S}_{\text{PKI}}$ and responds with $(r_{\text{PKI}i}, \rho_{\text{PKI}i}, \{d_{\text{PKI}i,j}\}_{1 \le j \le n, j \ne i})$

ii) On input the identity ID_i of *ID crypto system* participants U_{ID_i}
- If $\text{coin}_{sid_{\mathcal{U}_i}^{\chi}} = 0$ and $\text{coin}_i^{\mathcal{L}_1} = 0$
  a) Picks $\eta_{ID_i} \in \mathbb{Z}_q^*$ randomly and computes $r_{ID_i} = g^{\eta_{ID_i}}$
  b) Computes $\{d_{ID_i,j}\}_{1 \le j \le n}$ where $d_{ID_i,j} = sk_{ID_i} f_j^{\eta_{ID_i}}$
  c) Adds $(ID_i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{ID_i}, d_{ID_i,i})$ to $\mathcal{S}_{ID}$ and responds with $(r_{ID_i}, \rho_{ID_i}, \{d_{ID_i,j}\}_{1 \le j \le n, j \ne i})$
- If $\text{coin}_{\text{sid}_{\mathcal{U}_i}^{\chi}} = 0$ and $\text{coin}_i^{\mathcal{L}_1} = 1$
  a) Picks $\eta_{ID_i} \in \mathbb{Z}_q^*$ randomly and computes $r_{ID_i} = g^{\eta_{ID_i}} \prod_{l=1}^{n} y_{k-l+1}^{-1}$
  b) Computes $\{d_{ID_i,j}\}_{1 \le j \le n}$ where $d_{ID_i,j} = {r_{ID_i}}^{v_j} g^{o_i} y_j^{\eta_{ID_i}} \prod_{l=1}^{n, l \ne j} y_{k-l+1+j}^{-1}$
  c) Adds $(ID_i, \text{sid}_{\mathcal{U}_i}^{\chi}, \eta_{ID_i}, d_{ID_i,i})$ to $\mathcal{S}_{ID}$ and responds with $(r_{ID_i}, \rho_{ID_i}, \{d_{ID_i,j}\}_{1 \le j \le n, j \ne i})$

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* If $\mathrm{coin}_{sid^{\chi}_{\mathcal{U}_i}}=1$ and $\mathrm{coin}_i^{\mathcal{L}_1}=0$
  a) Picks $\eta_{IDi}\in\mathbb{Z}_q^*$ randomly and computes $r_{IDi}=g^{\eta_{IDi}}y_{k-i+1}$
  b) Computes $\{d_{IDi,j}\}_{1\le j\le n}$ where $d_{IDi,j}={r_{IDi}}^{\nu_j}g_1^{\mu_i}y_j^{\eta_{IDi}}y_{k-i+1+j}$
  c) Adds $(ID_i,\mathrm{sid}^{\chi}_{\mathcal{U}_i},\eta_{IDi},d_{IDi,i})$ to $\mathcal{S}_{ID}$ and responds with $(r_{IDi},\rho_{IDi},\{d_{IDi,j}\}_{1\le j\le n,j\ne i})$
* Else $\mathrm{coin}_{sid^{\chi}_{\mathcal{U}_i}}=1$ and $\mathrm{coin}_i^{\mathcal{L}_1}=1$
  a) Picks $\eta_{IDi}\in\mathbb{Z}_q^*$ randomly and computes $r_{IDi}=g^{\eta_{IDi}}\prod_{l=1}^{n,l\ne i}y_{k-l+1}^{-1}$
  b) Computes $\{d_{IDi,j}\}_{1\le j\le n}$ where $d_{IDi,j}={r_{IDi}}^{\nu_j}g_1^{\mu_i}y_j^{\eta_{IDi}}\prod_{l=1}^{n,l\notin\{i,j\}}y_{k-l+1+j}^{-1}$
  c) Adds $(ID_i,\mathrm{sid}^{\chi}_{\mathcal{U}_i},\eta_{IDi},d_{IDi,i})$ to $\mathcal{S}_{ID}$ and responds with $(r_{IDi},\rho_{IDi},\{d_{IDi,j}\}_{1\le j\le n,j\ne i})$

g) *Ek.Reveal* $(\Omega^{\chi}_{\mathcal{U}_i})$: Suppose $\sigma^{\chi}_{\mathcal{U}_i}=\{\sigma_1,\ldots,\sigma_n\}$, where $\sigma_l=(r_{\mathrm{PKI}/ID_l},\rho_{\mathrm{PKI}/ID_l},\{d_{\mathrm{PKI}/ID_{l,j}}\}_{1\le j\le n,j\ne l})$
   – If $\mathrm{state}^{\chi}_{\mathcal{U}_i}=$ successfully terminated, computes and responds with $\omega=\prod_{i=1}^{n_{\mathrm{PKI}}}r_{\mathrm{PKI}i}\prod_{i=1}^{n_{ID}}r_{IDi}$, $Q=\prod_{i=1}^{n_{\mathrm{PKI}}}pk_{\mathrm{PKI}i,j}\prod_{i=1}^{n_{ID}}pk_{ID_{i,j}}$
   – Else responds with null.

h) *Dk.Reveal* $(\Omega^{\chi}_{\mathcal{U}_i})$: $\mathcal{C}$ recovers $\mathrm{coin}_{\mathrm{sid}^{\chi}_{\mathcal{U}_i}}$ and related $\mathcal{U}_i^{\chi}$ first and then performs as follows.
   – If $\mathrm{coin}_{\mathrm{sid}^{\chi}_{\mathcal{U}_i}}=1$, abort.
   – Else if $\mathrm{state}^{\chi}_{\mathcal{U}_i}\ne$ successfully terminated, responds with null.
   – Else, recovers $S^{\mathrm{list}}_{\mathrm{PKI}/ID}$ and $\sigma^{\chi}_{\mathcal{U}_i}$ of $\mathcal{U}_i^{\chi}$ from S^{list}, where $\sigma_l=(r_{\mathrm{PKI}/ID_l},\rho_{\mathrm{PKI}/ID_l},\{d_{\mathrm{PKI}/ID_{l,j}}\}_{1\le j\le n,j\ne l})$, computes and responds with $dk_i=\prod_{j=1}^{n_{\mathrm{PKI}}}d_{\mathrm{PKI}j,i}\prod_{j=1}^{n_{ID}}d_{IDj,i}$

3) *Test* $(\Omega^{\chi}_{\mathcal{U}_i})$: $\mathcal{A}$ challenges with a fresh $\Omega^{\chi}_{\mathcal{U}_{\mathrm{PKI}/ID_i}}$ and two messages m_0,m_1. $\mathcal{C}$ performs
   a) Recovers all items $(ID_l^*,id_i^*,{pk_{ID_i}}^*,{sk_{ID_i}}^*,\mu_i,{\mathrm{coin}_i^{\mathcal{L}_1}}^*),(i,h_i,pk_{\mathrm{PKI}i},sk_{\mathrm{PKI}i},o_i,{\mathrm{coin}_i^{\mathcal{L}_{\mathrm{PKI}}}})$ from $\mathcal{L}_1$ and $\mathcal{L}_{\mathrm{PKI}}$.
   b) If $\mathrm{coin}_{\mathrm{sid}^{\chi}_{\mathcal{U}_i}}=1$, all ${\mathrm{coin}_i^{\mathcal{L}_{\mathrm{PKI}}}}^*=0$ and only one ${\mathrm{coin}_i^{\mathcal{L}_1}}^*=1$ exists, and $\{\sigma_{\mathrm{PKI}/ID_l}^*\}_{1\le j\le n,j\ne l}$ is not a forgery, continue. Else, abort.
   c) Recovers all items $\eta_{\mathrm{PKI}/ID_1}^*,\ldots,\eta_{\mathrm{PKI}/ID_n}^*$ from $\mathcal{S}^*_{\mathrm{PKI}}$ and $\mathcal{S}^*_{ID}$ and computes $ek^{\chi}_{U_i}=(w^*,Q^*)$ where $w^*=g^{\sum_{l=1}^{n}{\eta_{\mathrm{PKI}/ID_l}}^*}$, $Q^*=\hat{e}(g^{\sum_{l=1}^{n_{ID}}\mu_l^*}y_k,g_1)\cdot\hat{e}(g^{\sum_{l=1}^{n_{\mathrm{PKI}}}o_l^*},g)$.
   d) Computes $c_1=\varpi^*$, $c_2=\varpi^{\sum_{l=1}^{n}{\eta_{\mathrm{PKI}/ID_l}}^*}$, selects $\vartheta\in\{0,1\}^{\varsigma}$ randomly, and computes $c_3=m_b\oplus\vartheta$, where $b\in\{0,1\}$.
   e) Responds with $c=(c_1,c_2,c_3)$.

4) *Response*: Eventually, once $\mathcal{C}$ receives $b'\in\{0,1\}$ from $\mathcal{A}$, $\mathcal{C}$ could solves and responds with the k-BDHE problem as $\Phi_i\cdot\hat{e}(g_1^{-\sum_{l=1}^{n_{ID}}\mu_l^*}g^{-\sum_{l=1}^{n_{\mathrm{PKI}}}o_l^*},\varpi)$ where (Φ_i,Ψ_i) is randomly picked from $\mathcal{L}_3$.

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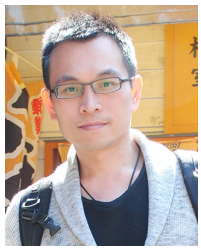
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